

F78DS Coursework 2 Specification

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Due date: Tuesday 7th April 2026 - 10:00 am UK = 5:00 pm Malaysia time.

Data

The data is provided in a single comma separated (CSV) file named “[F78DS-Essay-Features.csv](#)”. This data was derived from a set of essays and is used to describe the essay features in numeric information. The columns in this file are:

- **essayid** - a unique id to identify the essay
- **chars** - number of characters in the essay, including spaces
- **words** - number of words in the essay
- **commas** - number of commas in the essay
- **apostrophes** - number of apostrophes in the essay
- **punctuations** - number of punctuations (other than commas, apostrophes, period, questions marks in the essay)
- **sentences** - number of sentences in the essay, determined by the period (fullstops)
- **questions** - number of questions in the essay, determined by the question marks
- **POS** - total number of Part-of-Speech discovered
- **prompt_words** - words that are related to the essay topic
- **synonym_words** - words that are synonymous
- **unstemmed** - number of words that were not stemmed in the essay
- **stemmed** - number of words that were stemmed (cut to the based word) in the essay
- **score** - the rating grade, ranging from 1 – 6

Note that this data is a pre-processed data on a set of essays that were provided on [Kaggle](#) [↗\(https://www.kaggle.com/c/asap-aes\)](https://www.kaggle.com/c/asap-aes). **DO NOT** download or process/wrangle the data from the original source.

The Hewlett Foundation: Automated Essay Scoring

Develop an automated scoring algorithm for student-written essays.

Overview

Data

Code

Models

Discussion

Leaderboard

Rules

Overview

Start

Feb 10, 2012

Close

May 1, 2012

Prizes & Awards

\$100,000
Awards Points & Medals

Participation

195 Competitors
153 Teams
2,493 Entries

For the coursework, you are given a single file “[F78DS-Essay-Features.csv](#)”. The description of the data is as described above. Do use the Canvas Discussion to clarify further on the fields that you are unsure, e.g., “*prompt*”.

Submission

There are 2 types of submission for this coursework;

- Canvas Submission
- Kaggle Submission

Canvas Submission

Similarly to Coursework 1, this coursework has to be done using the **Jupyter Notebook** only. Your Jupyter Notebook has to use the Markdown language for proper formatting of the report and answers, with inline Python code and graphs.

You are to hand in **two** files on Canvas:

1. The **Jupyter Notebook file (.ipynb)** that contains a working copy of your report (using Markdown) and Python code that answers the questions.
2. A **PDF** file that is generated from your Jupyter Notebook. On your Jupyter Notebook page, go to the Menu item “Kernel” and then select “Restart & Run All”. Make sure it can be executed and no errors. Then download it as a PDF document. If you do not have the “Download as” with the PDF version, you can do a “Print Preview”, then “Print” the document, and then select “Save as PDF”. Note that there are other ways to do this, depending on the environment that you are in. Alternatively, you can download as HTML and then “Print” that to PDF.

Kaggle Submission

The purpose of the Kaggle submission is to provide you with an introductory experience on how machine learning models are evaluated. Data (“[F78DS-Essay-Features.csv](#)”) will be provided on the Kaggle site for building the model. **Here is the link to the [Kaggle HWU F78DS January Semester 2026](#)** [↗\(https://www.kaggle.com/t/7cda29f6b7fb4c9595e1a68c2935096a\)](https://www.kaggle.com/t/7cda29f6b7fb4c9595e1a68c2935096a).

Another file, called the “[F78DS-Essay-Features-Submission.csv](#)” consists of data where there are no labels (no ‘[score](#)’ column). The whole purpose is to be able to predict those labels for this data set. You are to output the data to a CSV file that contains 199 rows (200 if include the headers) and 2 columns, the column “[essayid](#)” and another column named “[score](#)”. A sample file without the ‘[score](#)’ entries is also available “[H00999999-YourName-1.csv](#)” (this is to give an idea of what is expected).

Competition Submission and Evaluation Method

When you logged on the competition page, submissions can be made on the second rightmost tab (this may move).

You will be asked to read and understand the rules of the competition. When you have prepared your submission CSV file, drag and drop your CSV file (please name it using your “HWID-name-version.csv” into the space allocated and give it a description, so that you can track it. Please note that the submission process may take a **LONG** time, so please be patient. The system is evaluating your submission. Also, you **can only submit it a few times in a day**, so make sure you plan and try it out early.

The competition uses the QWK evaluation method, which will result in a score of between 0.0 and 1.0. (Generally, score between 0.6 – 0.8 is good and above 0.8 is considered matching).

Clarifications

This coursework is not meant to provide step by step instructions and we expect to have some clarification questions. The questions can be as simple as “What is a PDF file” to something relating to the possible meaning of a column in the CSV file.

I would like you to post these questions on “[Canvas Discussions](https://canvas.hw.ac.uk/courses/32191/discussion_topics/244748) (https://canvas.hw.ac.uk/courses/32191/discussion_topics/244748)” and I strongly encourage interactions between all of you there. Some of the questions probably don’t have a single answer or a correct answer and is up to each individual’s interpretation. Just make sure that you **do not post coding solutions** in the forum. We will close the discussion board at 10:00 am UK on Monday 6th April.

Coursework

This coursework is worth 30 marks, which makes up for 30% of this course’s assessment. This coursework has to be done using the **Python programming** language in the **Jupyter Notebook environment**. It should also be formatted properly using the Markdown language. Below is an example from past submission (that’s totally unrelated to this Coursework).

Example 1

This example has a code cell, the output, which is a rather nice pie chart (with some labels that aren’t ideal) and a short explanation

Tasks

You should start your coursework by providing the title of the coursework and course code, your name and student ID, e.g.

Example 2

Same as your Coursework 1 format

Thereafter, you are to complete the following tasks (note that any reference to the library/function may not be accurate, so you will need to discover what is necessary):

1. Introduction

1. Start with an introduction to the coursework.
2. Importing the necessary libraries, read the file ('`F78DS-Essay-Features.csv`'), and provide some description of the data you have read (you do not need to repeat the description given in this file for each field).
3. The data has been pre-processed but you are expected to check the data to ensure that it is in a Tidy Data format.

2. Supervised Learning

1. Explain supervised machine learning, the notion of labelled data, and the training and test datasets.
2. Separate the features and the label (Hint: the label in this case is the '`score`').
3. Conduct feature selection or feature engineering (if appropriate) to improve the model performance.
4. Use the `sklearn.model_selection.train_test_split` function to split your data for training and testing. Explain the parameters used in the function, especially the sampling method.

3. Classification

1. Explain the difference between binary and multi-class classification.
2. In preparation for Decision Tree (DT) classification, your data may need to be normalised.
 1. Describe what you understand from this need to normalise data (this is in your Week 7 laboratory).
 2. Choose and use the appropriate normalisation functions available (if needed) from `sklearn.preprocessing` and scale the data appropriately.
3. Use the Decision Tree (DT) algorithm to build the model.
 1. Describe the Decision Tree model that is used, and the selected options that are appropriate in this exercise.
 2. Explain the hyperparameters (if any) tuning that you did and from your understanding, why the value chosen is appropriate for this coursework.
 3. Write the code to build the model using your training dataset.

4. Predict

1. Using the testing dataset you created in 2.4 above, conduct the prediction for the '`score`' (label).
2. Display the confusion matrix (it should look like a 6x6 matrix). Unlike the lectures, where it is just a 2x2, you are now introduced to a multi-class classification.
3. Explain Quadratic Weighted Kappa (QWK). This is not in your lectures.
4. Use the `sklearn.metrics` library to code and obtain the QWK score.
5. Use a more complex algorithm, Random Forest, to build an alternative model (optional).
 1. Conduct steps 3.3 and 3.4 above to obtain an alternative model's results.
 2. Briefly explain why this alternate model performs worse or better than the Decision Tree model.

4. Kaggle Submission

1. Read the 'F78DS-Essay-Features-Submission.csv' file and use the model you built earlier (whichever model that you think is better) to predict the 'score'.
2. Unlike the previous section, you have a testing (also sometimes referred to as the validation) dataset where you know the 'score' and will be able to test for the accuracy. In this part, you don't have a 'score' and you have to predict it and submit it to the competition site. (To build a more robust model, you can do cross-validation. This in reference back to 2.4 above where you are to split your data for training and testing - will leave it to you to learn about this on your own).
3. Output your prediction to a CSV file that contains 2 columns, 'essayid' and 'score'. It should have a total of 200 lines (1 header, and 199 entries).

5. Conclusion

1. Conclude your coursework.

Have Fun!

Upon completion of this coursework, you should have some experience with the *Collect*, *Wrangle*, *Analyse* and *Present* process that is core to the role of a Data Scientist (See Lecture 1).

Congratulations!

By completing Coursework 1, you would have experienced looking, understanding, and auditing data. You would also have provided exploratory analytics using descriptive statistics and some visualisation. In doing so, you would have had to spend some time sieving through the data to understand it. That was the intention to get you to experience it.

For Coursework 2, we skipped the data wrangling (it was already done earlier for you) and moved to focus on preparing your data for analytics, conducting machine learning using available libraries to build various models, output your results and got the results to be independently evaluated. In Coursework 2, you should also have experienced (beyond the lectures):

- Hyperparameter tuning for Decision Trees (DT), and/or other supervised machine learning algorithms.
- A different measurement metrics, a Quadratic Kappa Score, which is a variation of the Cohen Kappa Score.
- Created a Kaggle account, navigate through the Kaggle site, and understand a basic submission for competition evaluation.

You should now be ready to start to build a machine learning portfolio by entering proper Kaggle competitions. This should give you an introduction to the role of a data scientist